

Practice Questions

CS475/CS675 - Computer Graphics

September 2, 2022

1 Rasterization

1. Give an algorithm to rasterize a circle of radius r that is centered at the origin. Give an analysis of the error your rasterization algorithm incurs while choosing which pixels to color. Also show whether your algorithm produces a rasterization with least deviation (or error) from the true circle given by $x^2 + y^2 = r^2$.

2 Clipping

1. A rectangular clipping window is defined by the following window coordinates: $(0, 0)$ for the left, bottom corner and $(5, 4)$ for the right, top corner. We are also given two line segments: Line AB (from $A(-1, -1)$ to $B(6, 6)$) and Line CD (from $C(-1, 1)$ to $D(4, -3)$) that we want to clip against the window using the Cohen-Sutherland Clipping algorithm. What is the sequence of bitcodes generated by the algorithm when it is run on the lines AB and CD . Also, mention what is the final result of the clipping.

The 4 bit bitcode PQRS are defined as per the following convention:

P is 1 if $x < 0$
 Q is 1 if $x > 5$
 R is 1 if $y < 0$
 S is 1 if $y > 4$

2. Does the Cyrus-Beck clipping algorithm have any advantage over the Cohen-Sutherland algorithm? Explain with an example.

3 Transformations

1. Two lines, $AB \equiv -(2/3)x + y = -(1/3)$ and $EF \equiv x + y = 1$ are transformed using the following transformation matrix,

$$T = \begin{bmatrix} 1 & 1 & 0 \\ 2 & -3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

What is the point of intersection of the transformed lines?

2. A triangle ABC formed by the vertices $A(4, 1)$, $B(5, 2)$, $C(4, 3)$. What are the final coordinates of the vertices if the triangle is first reflected about the x -axis and then reflected about the line $y = -x$. Show that this transformed configuration of vertices is also obtainable by rotating the original triangle about the origin by 270 degrees.
3. A cuboid has sides of length 4 units, 6 units and 10 units respectively. Write an anisotropic 3D scaling transformation, T , to transform this cuboid to a cube of side 2 units. What does the transformation do to the volume of the cuboid?
4. If $T(P) = MP + t$ where M is a general 2×2 matrix and t and P are 2-dimensional vectors, is T a linear map or not? Prove your claim.

4 Viewing

1. Given a set of arbitrarily oriented line segments in 3D space, which of the line segments will retain their original lengths under a perspective projection on the $z = 0$ plane through a camera center lying on the positive z -axis?
2. Given two line segments AB and CD in 3D space, prove that their point of intersection maps to the point of intersection of $A'B'$ and $C'D'$ where the points A', B', C', D' are the perspective projections of points A, B, C, D respectively.
3. How can we get the following in OpenGL (answer in terms of a general sequence of operations, exact syntax of commands is not necessary):
 - (a) Axonometric Projections
 - (b) Oblique Projections
 - (c) 2 and 3 point perspective projections

5 The Modelling-Viewing Pipeline

1. Do we need to have a modelling-viewing pipeline? Can we, instead just form the requisite perspective projection matrix and multiply it with all objects to form the image? Answer keeping in mind that the process has to be implemented and is not purely based on mathematical/theoretical considerations.
2. Why do we do Projection Normalization in our Modelling-Viewing Pipeline?
3. Where should clipping be done in the modelling-viewing pipeline and why?
4. How can we get the following in OpenGL (answer in terms of a general sequence of operations, exact syntax of commands in not necessary):
 - (a) Axonometric Projections
 - (b) Oblique Projections
 - (c) 2 and 3 point perspective projections
5. Why is the Z (or depth) coordinate carried through the Modelling-Viewing pipeline when we only need the X and Y coordinates to locate pixels on the final image?
6. What kind of transformation does the Z coordinate undergo from the WCS to VCS to CCS to NDCS to DCS? Answer in terms of changes in depths that you can resolve successfully under such a sequence of transformations?
7. What are Euler angles? How can we define a viewing transformation in terms of Euler angles? To read up on Euler angles, check the resources linked to the course page. The link to the resources page is given on Moodle too as “CS475/CS675 Online Resources.”